

Alg2 Notes 6-8: Graphing Radical Functions

Warm-Up:

Given $f(x) = 3x + 2$

$g(x) = 4 - 5x$

$h(x) = \sqrt{x+2}$

1. $(f - g)(-2)$

2. $\left(\frac{h}{g}\right)(-1)$

3. $(f \circ g)(-1)$

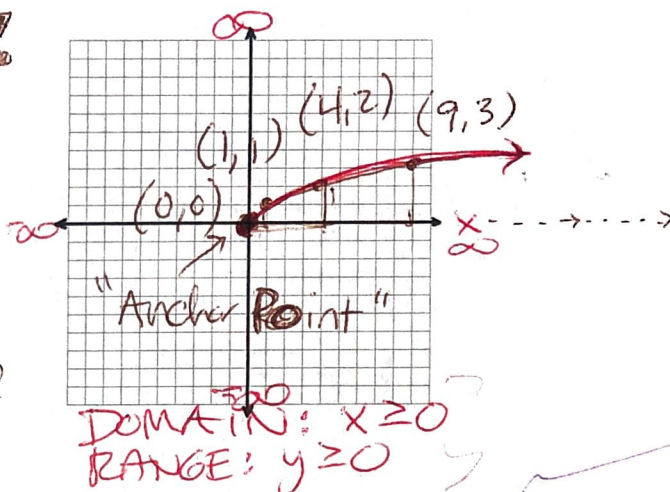
The Parent Square Root Function

$f(x) = \sqrt{x}$ set of Real #'s

Typically, pick x values $-2, -1, 0, 1, 2$
To discover its shape let's plot points this way

x	$\sqrt{x}$
-2	0
-1	0
0	0
1	1
4	2
9	3

Instead, let's
use #'s for x
that we know
the sq root of



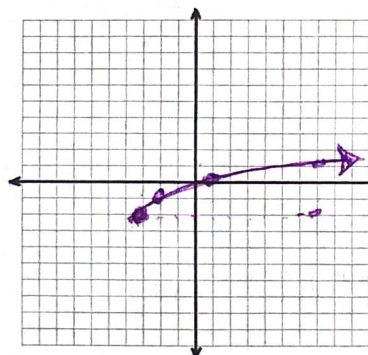
Transforming the Square Root Function

$f(x) = a\sqrt{x-h} + k$

- a moves us up or down
- h move us left or right
- k moves us up or down

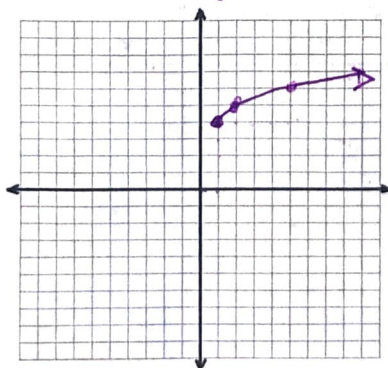
if a is negative, graph flips over x -axis

if h is negative, graph flips over y -axis

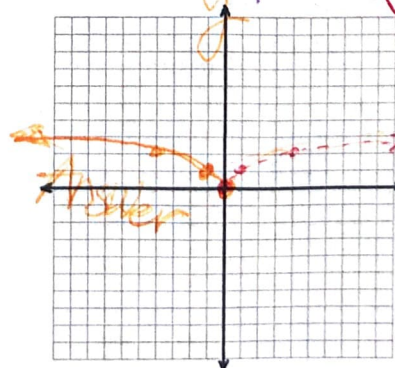


$y = \sqrt{x+3} - 2$
left 3
Down 2
Get Anchor point!

Ex. Graph $y = \sqrt{x-1} + 4$
right one - up 4



Ex. Graph $y = \sqrt{-x}$



flip graph over y axis

→ We can find $\sqrt[3]{}$ of neg. #'s!

The Parent Cube Root Function

$$f(x) = \sqrt[3]{x}$$

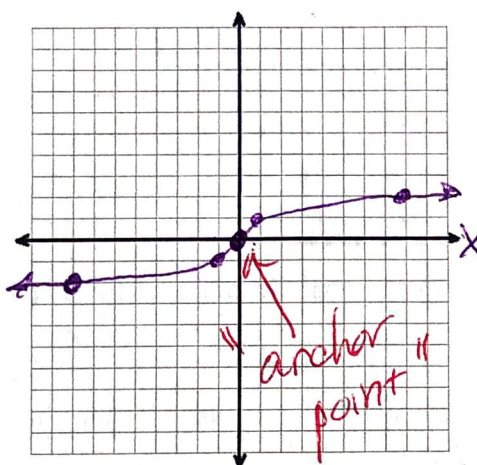
To discover its shape let's plot points this once

x	$\sqrt[3]{x}$
-8	-2
-1	-1
0	0
1	1
8	2

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$



Transforming the Cube Root Function

ref. @
x-axis

$$f(x) = a\sqrt[3]{x-h} + k$$

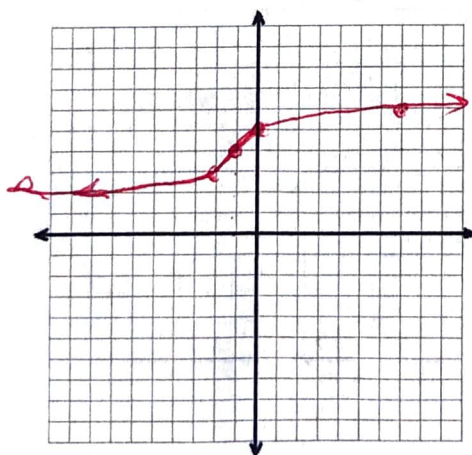
$$y = -\sqrt[3]{x} - 2$$

Still follow PEMDAS with
our transformations

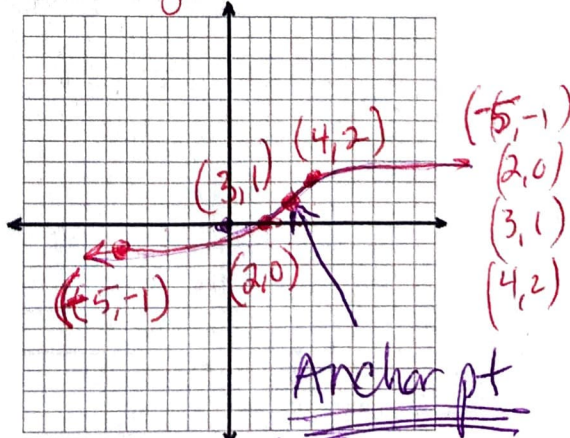
You

Ex. Graph $y = \sqrt[3]{x+1} + 4$

left 1 up 4



~~Graph $y = (x-3)^2$~~ Write eqn
for graph.



3 units right, 1 unit up

$$y = \sqrt[3]{x-3} + 1$$